Name
Date
Course

Geometry Final Test (Part I)

I) Write your name, the date, and the course title (i.e. Geometry).
II) Show all work.
III) If a problem requires algebraic steps to solve, draw a rectangle around your final answer to distinguish it from the other work.
IV) You may use a calculator for some problems on this test.
V) Each problem is worth four points for a total of 100 points.

① Match a, b, or c for parts i-iii below.

Original: If they sell beef jerky, then I will buy some for Jeff.

i) If I do not buy some beef jerky for Jeff, then they do not sell beef jerky.

ii) If I buy some beef jerky for Jeff, then they sell beef jerky.

iii) If they do not sell beef jerky, then I will not buy some for Jeff.

a) Converse

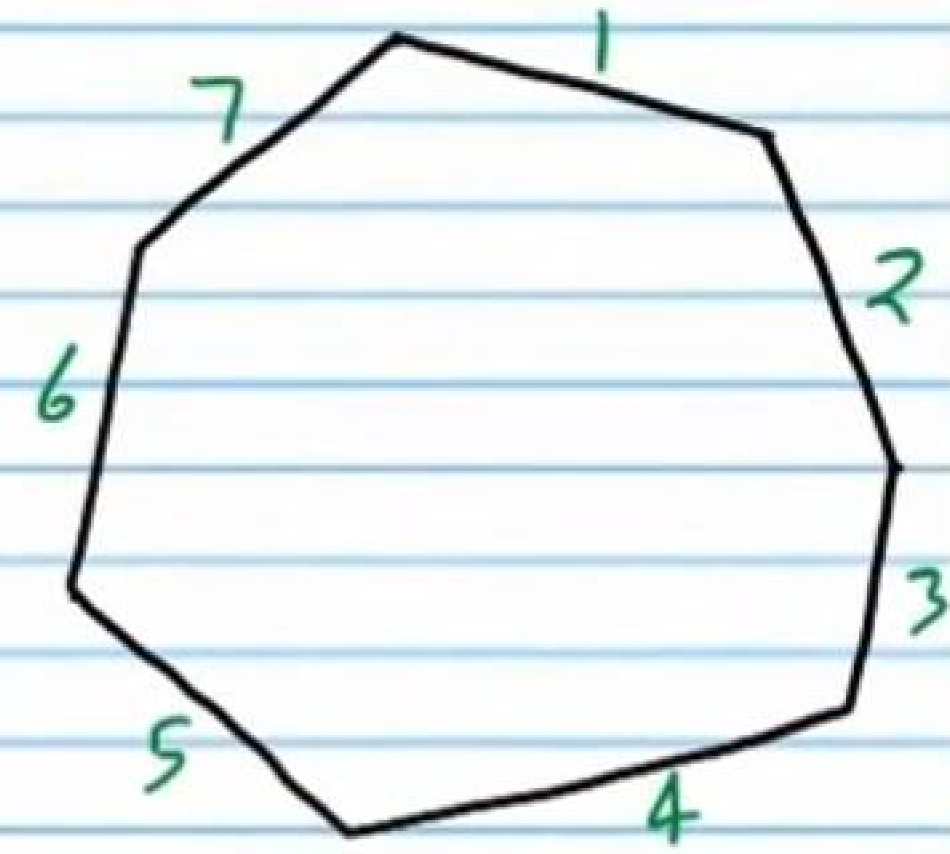
b) Contrapositive

c) inverse

iV) Identify the hypothesis and conclusion of the original statement.

② What is formed by the intersection of two planes?

③ i) Name the following polygon by its number of sides.

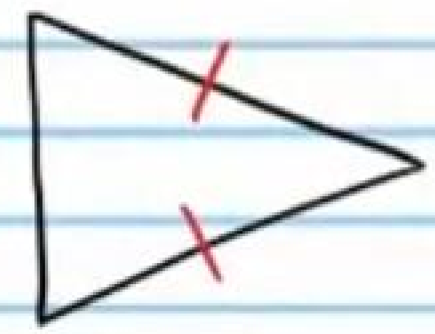


ii) Is it equilateral?

iii) Is it equiangular?

iV) Is it convex or concave?

④ i) If the triangle below were classified by its sides, what word would be used to describe it?



ii) If the triangle above were classified by its angles, what word would be used to describe it?

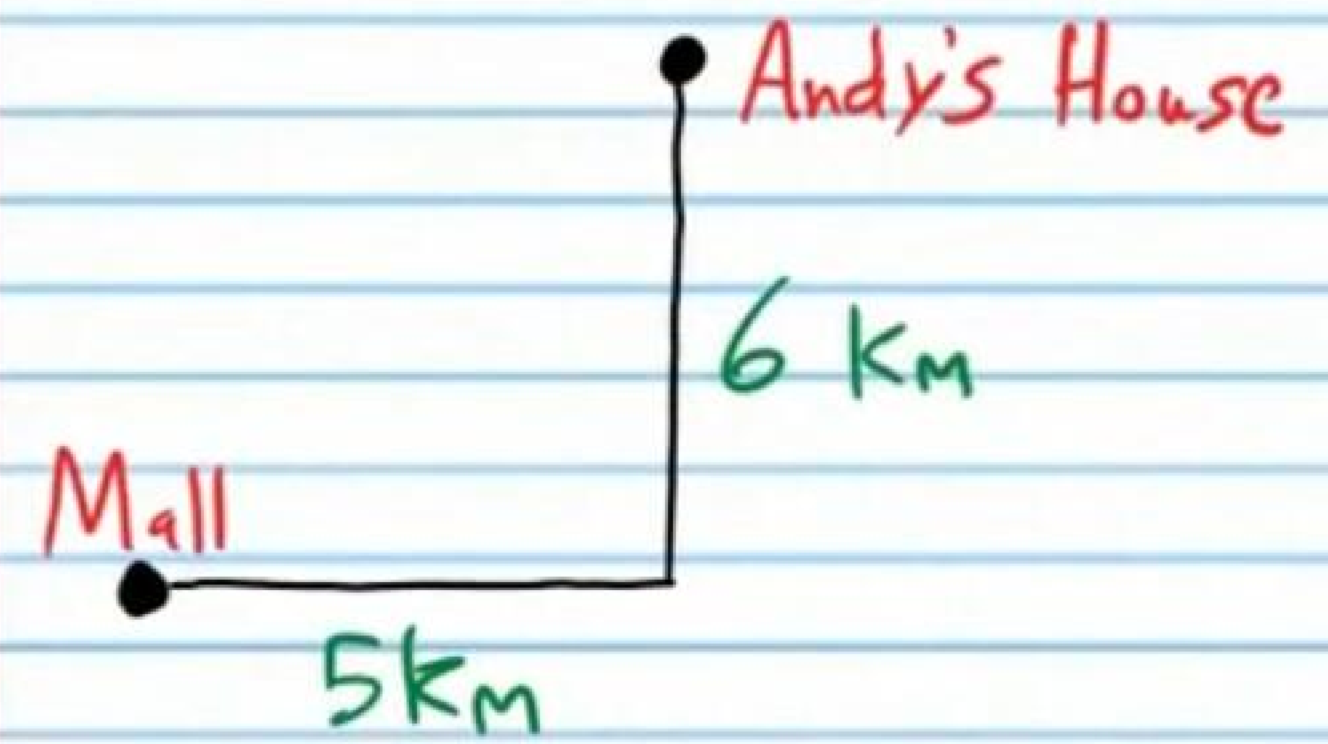
⑤ A conclusion based on logic, definitions, postulates, and theorems is produced using what type of reasoning?

⑥ Using the following information, what law can be used to form a conclusion?

a) If the weather is warm, then I can ride my bike.
b) If I ride my bike, then I will have fun today.

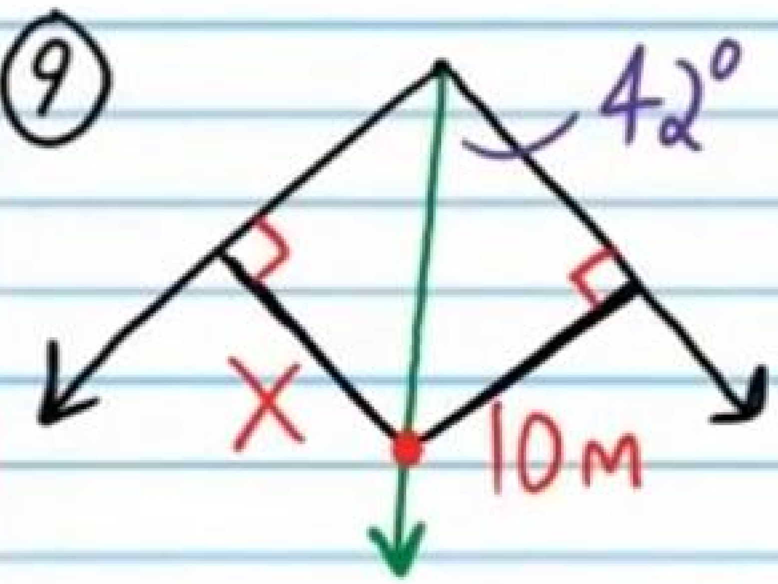
⑦ "Children must ride the bus to get to school." Write a counterexample for the previous statement.

⑧ What is the shortest distance from Andy's house to the mall? You may write your answer as an exact number or round to the nearest thousandth.

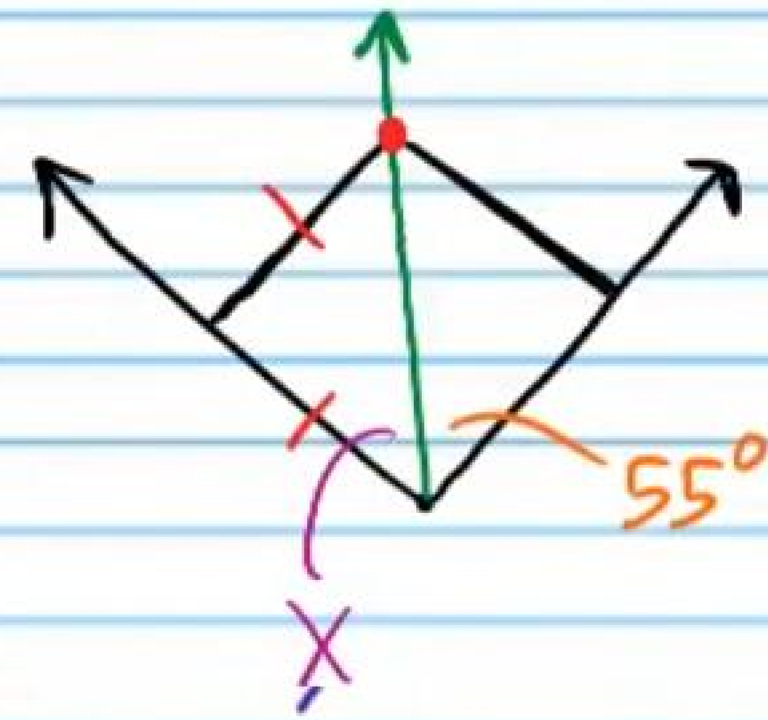


Find x if possible.

⑨



⑩



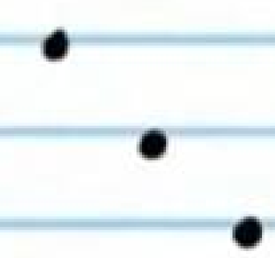
⑪ i) Predict the next number in the sequence below.

3, 30, 300, ...

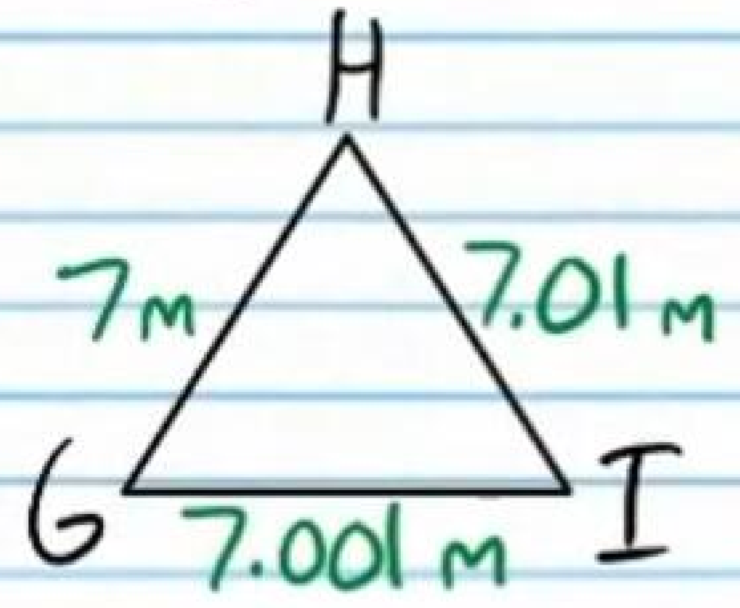
ii) What type of reasoning did you use to make this prediction?

iii) What is the special name we use for the prediction produced by this type of reasoning?

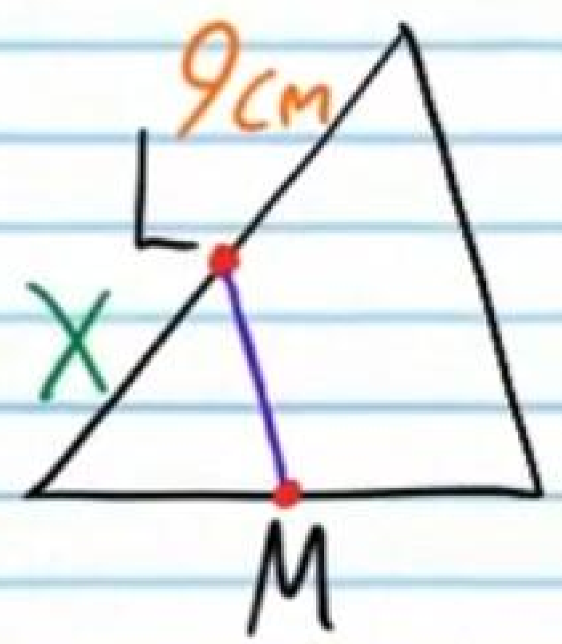
iv) Are the points below collinear? Explain.



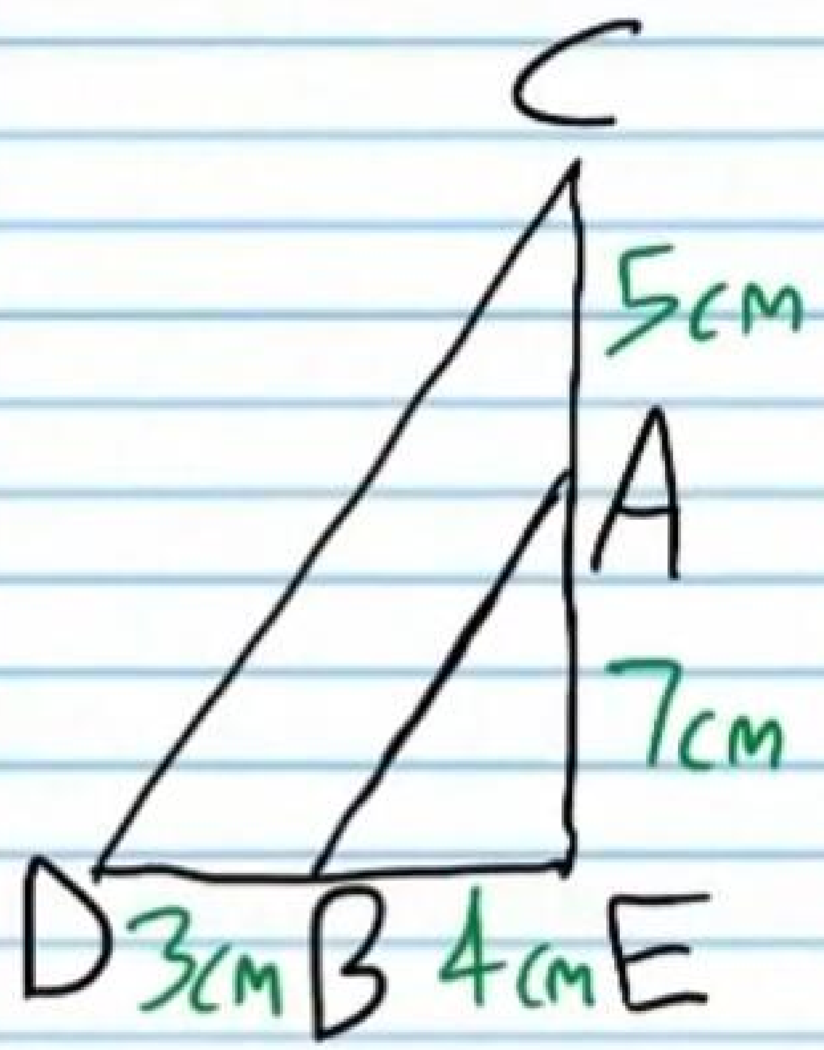
⑫ List the angles of the triangle below from least to greatest.



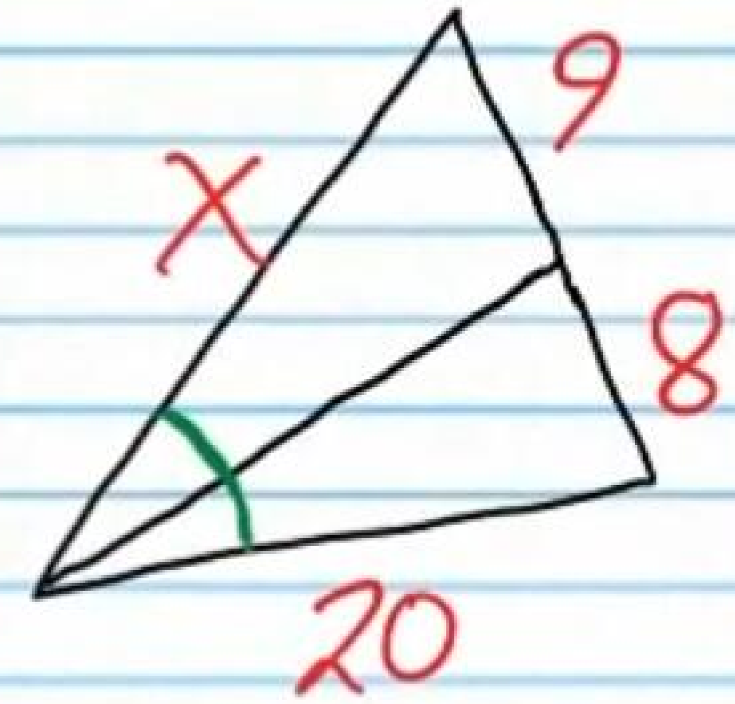
⑬ If $\overline{LM}$ is a midsegment of the triangle below, find X.



⑭ Is AB || CD? Show the proportions that prove your answer.



⑮ Find X.



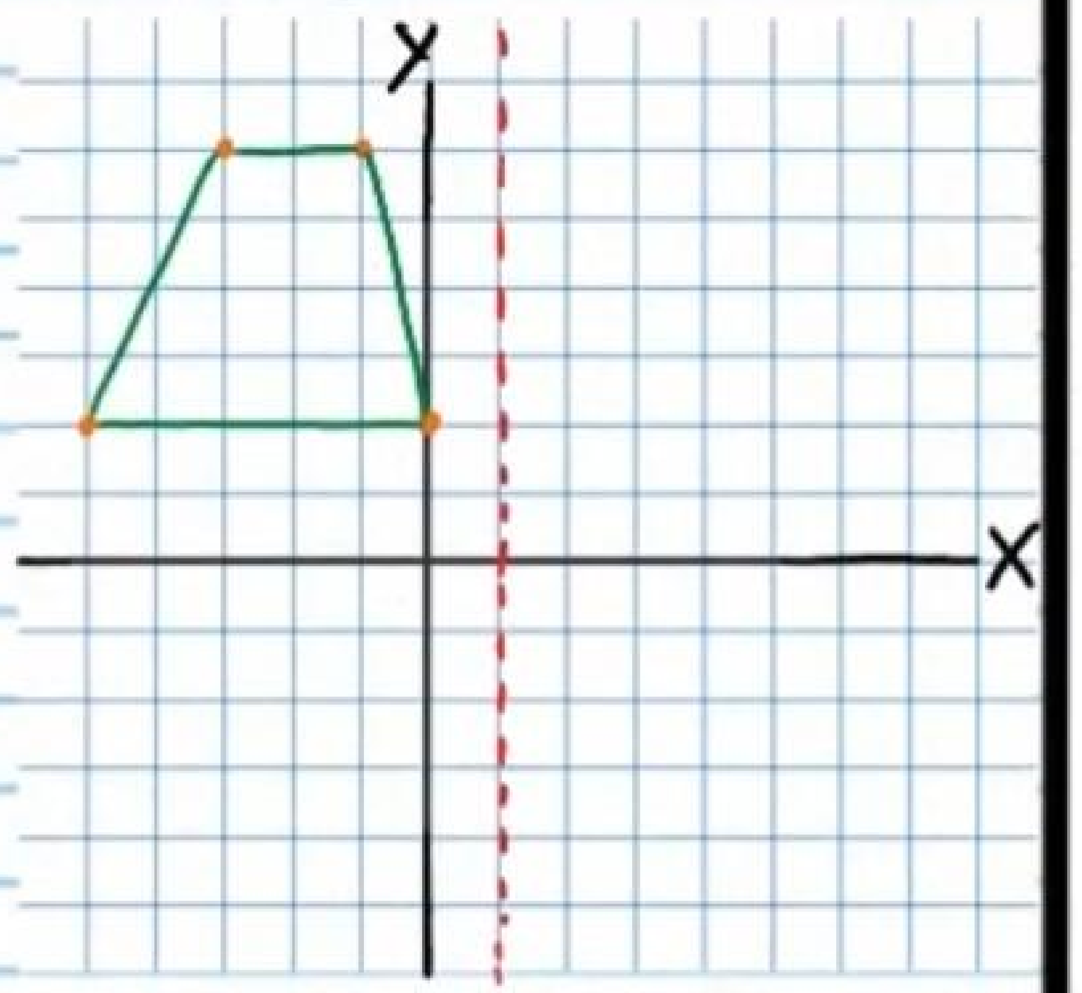
⑯ If the numbers below represent side lengths of a triangle, write an inequality that shows the possible lengths of the third side.

8, 9

⑰ Write the following statement in biconditional form.

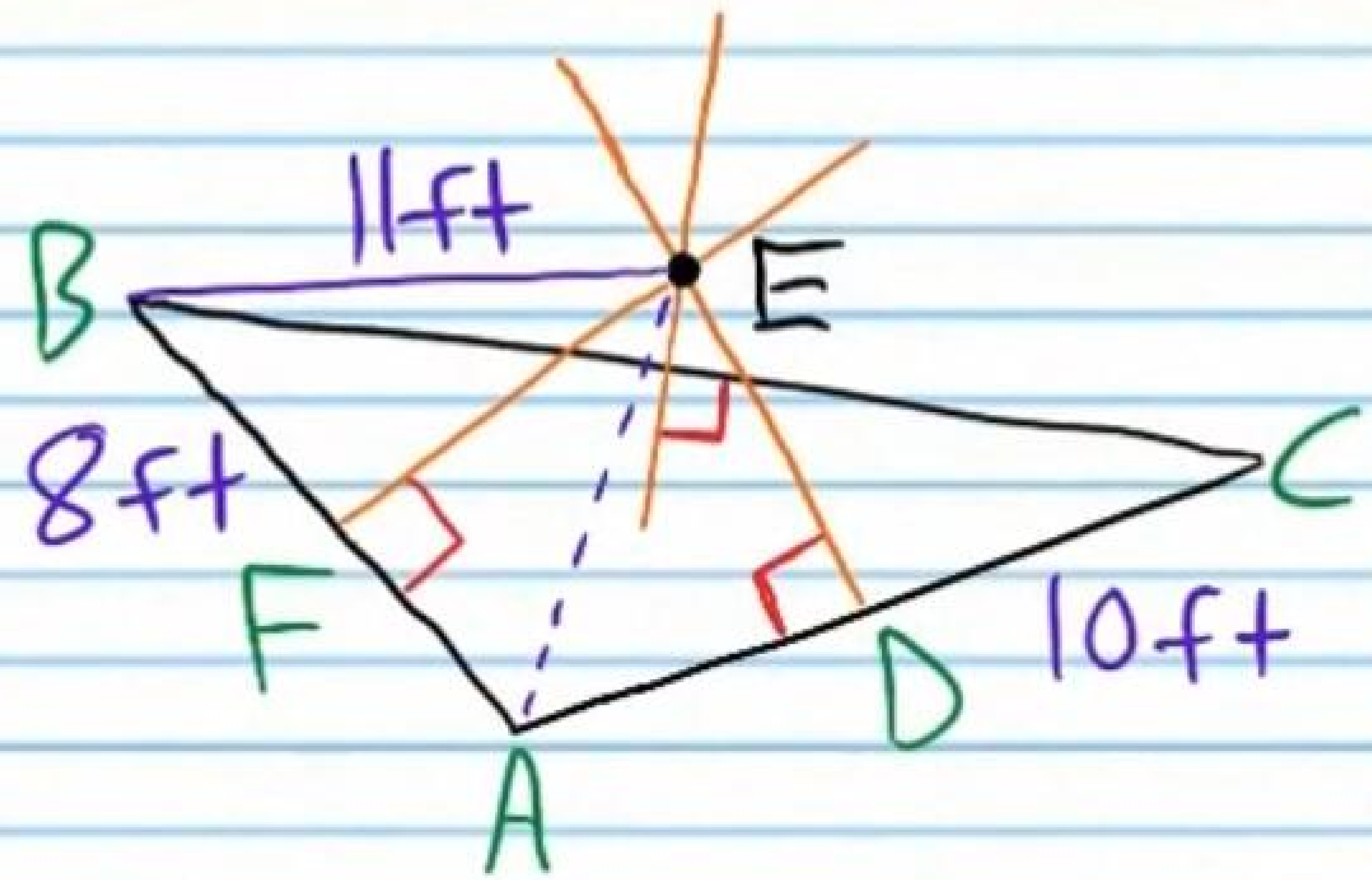
If an animal is a dog, then it has four legs.

(18) Given the preimage, graph the image.
Reflection about the dotted line



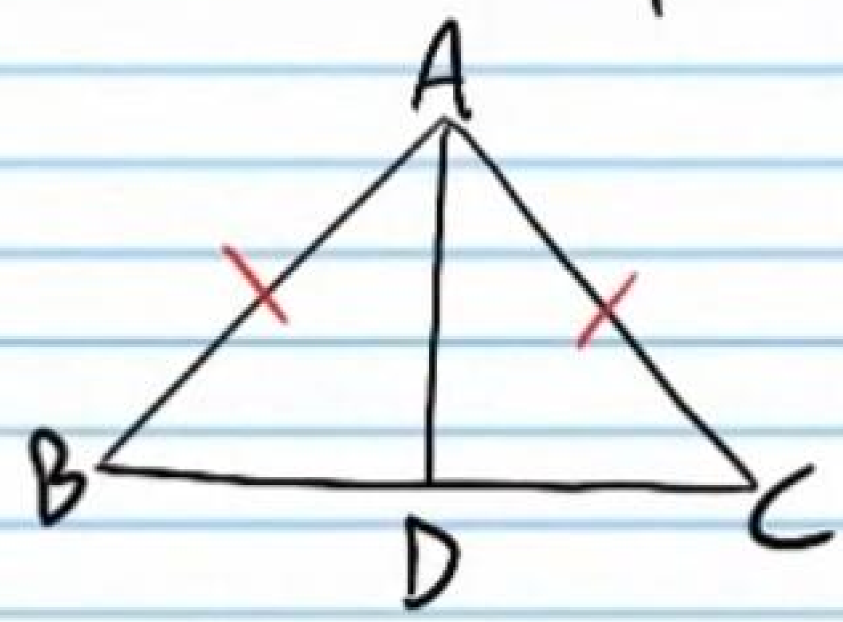
(19) If point E is the circumcenter of the triangle below, find the quantities listed.

i) AD
ii) BA
iii) AE
iv) EC

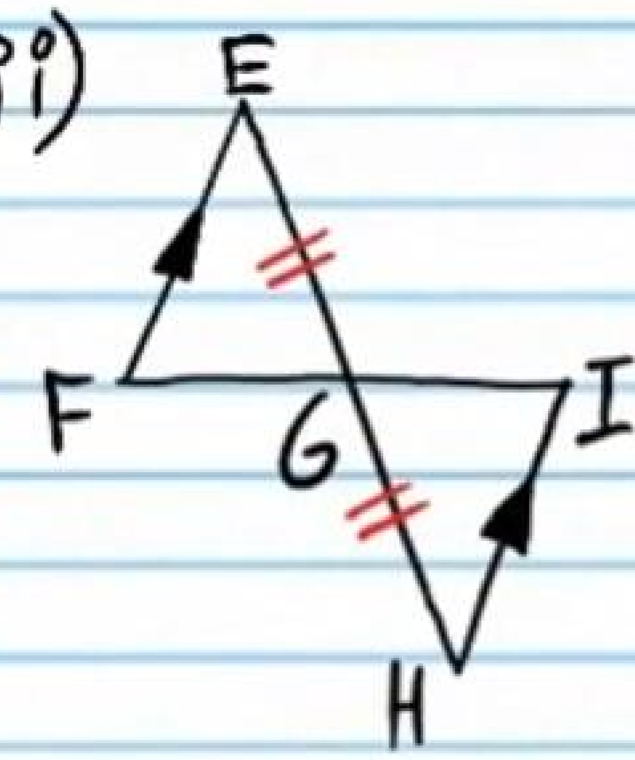


(20) In the problems below, determine if the triangles are congruent. If they are congruent, write a proper congruence statement and the triangle postulate or theorem that proves that they are congruent. You do not have to write a formal proof (i.e. a two-column proof).

i) D is the midpoint of $\overline{BC}$.



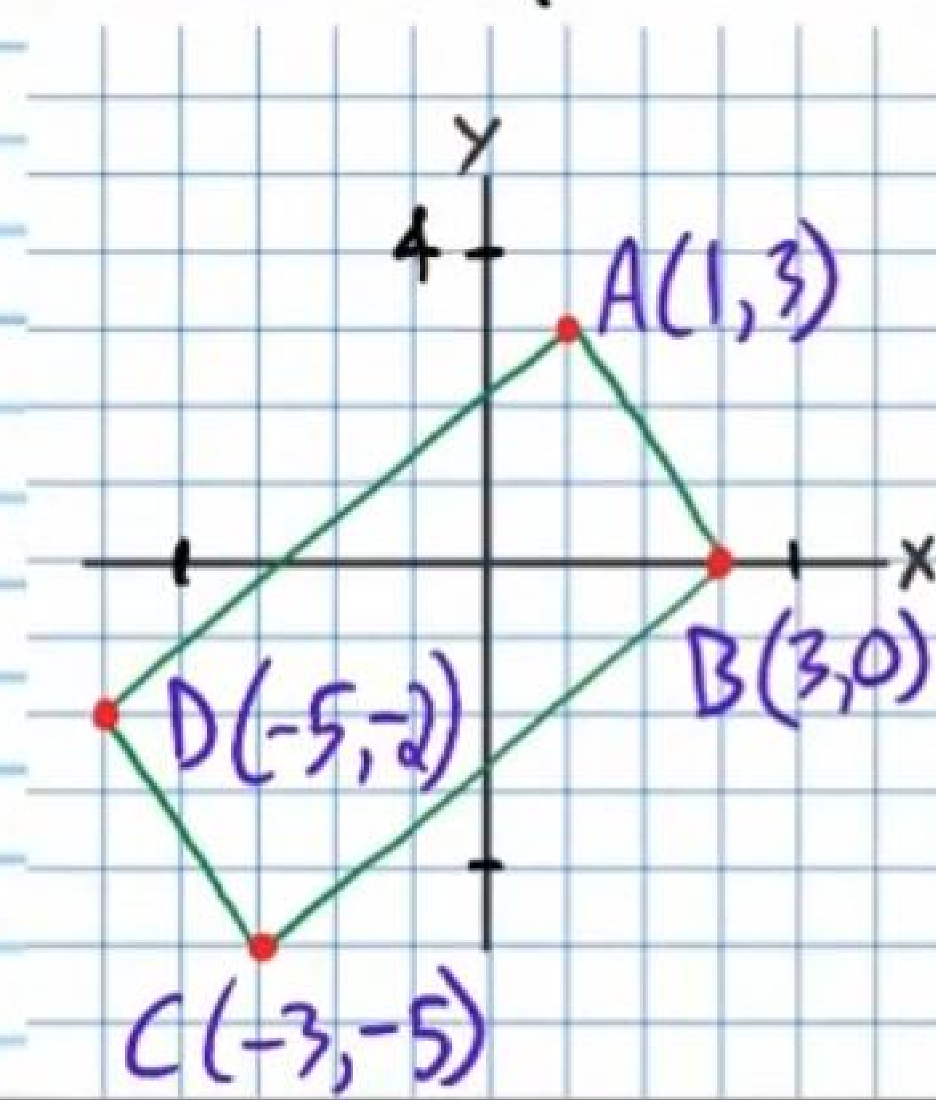
ii)



$\triangle ABD \cong$ ___?___

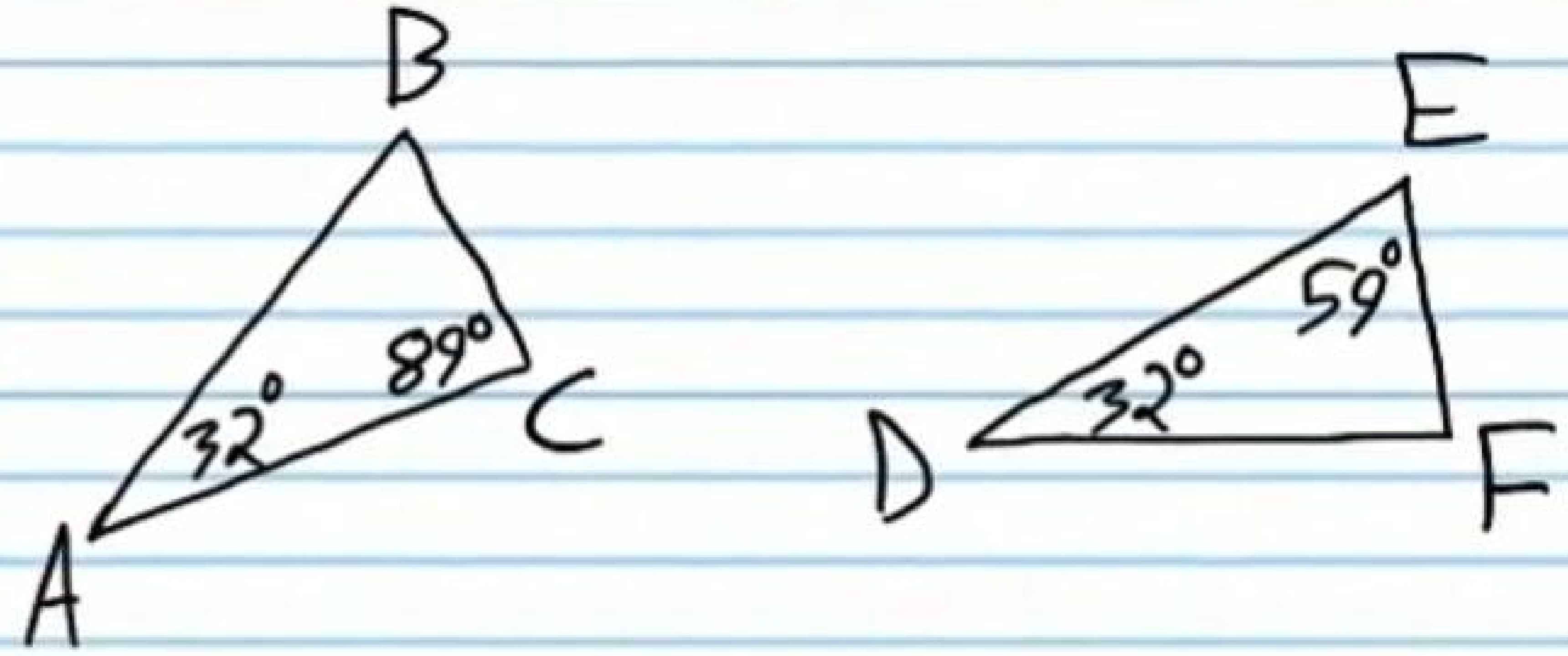
$\triangle FEG \cong$ ___?___

(21) Show that the diagonals of the quadrilateral below bisect each other.



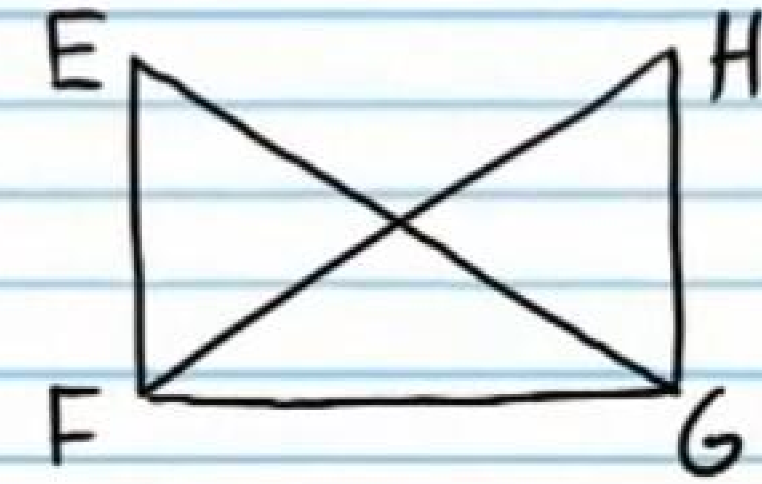
Show that the following statements are true using two-column proofs. You might not use the same number of steps written in Roman numerals. You may use more or less.

(22) ΔABC is similar to ΔDFE.



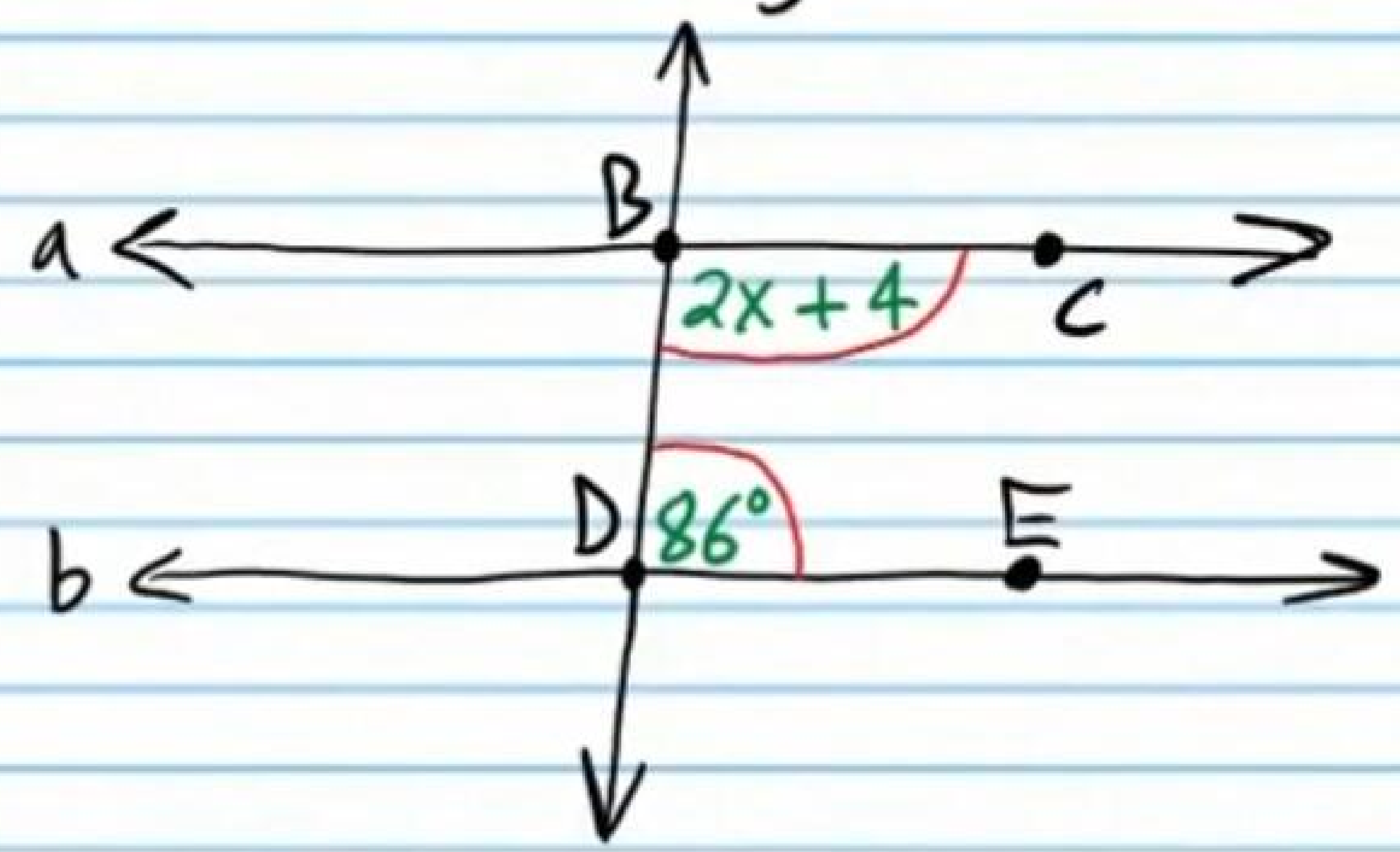
Statement	Reason
I)	
II)	
III)	
IV)	
V)	
VI)	
VII)	

(23) If $\overline{EF} \perp \overline{FG}$, $\overline{FG} \perp \overline{HG}$, and $\overline{EG} \cong \overline{FH}$, then $\angle E \cong \angle H$.



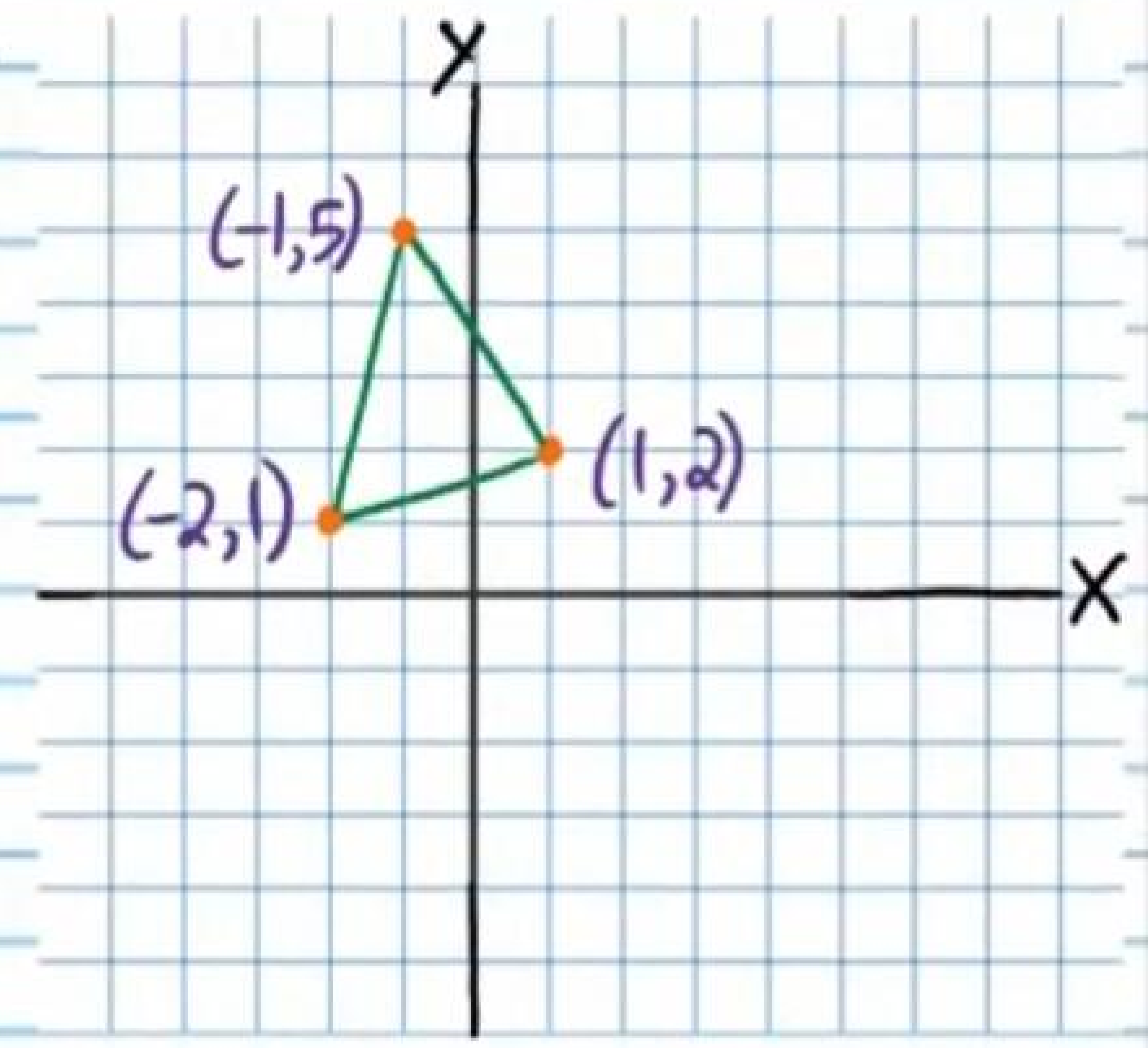
Statement	Reason
I)	
II)	
III)	
IV)	
V)	
VI)	
VII)	

(24) If $x = 45$ in the figure below, then $a \parallel b$.



Statement	Reason
I)	
II)	
III)	
IV)	
V)	
VI)	
VII)	
VIII)	
IX)	

(25) Find the coordinates of the orthocenter of the triangle below.



Geometry Final Test (Part I) Answers

1. i) $\boxed{b}$ ii) $\boxed{a}$ iii) $\boxed{C}$

iV) Hypothesis: They sell beef jerky.
Conclusion: I will buy some for Jeff.

2. $\boxed{\text{A line.}}$

3. i) $\boxed{\text{Heptagon}}$ or $\boxed{\text{Septagon}}$ ii) $\boxed{\text{No}}$ iii) $\boxed{\text{No}}$ iV) $\boxed{\text{Convex}}$

4. i) $\boxed{\text{Isosceles}}$

ii) $\boxed{\text{Acute}}$

5. $\boxed{\text{Deductive}}$

6. $\boxed{\text{Law of Syllogism}}$

7. Any of the following are acceptable: Parents can drive their children, Kids can ride their bikes, the subway, train, horses, fly a plane, walk, or use a scooter, skateboard, or motorcycle. Other examples are acceptable.

8. $\boxed{\sqrt{61}\text{ Km}}$ or $\boxed{7.81\text{ Km}}$

9. $\boxed{\text{Not enough information.}}$

10. $\boxed{\text{Not enough information.}}$

11. i) $\boxed{3{,}000}$ ii) $\boxed{\text{Inductive}}$ iii) $\boxed{\text{Conjecture}}$

iV) $\boxed{\text{Yes. One line can be drawn through all three points.}}$

12. $\boxed{\angle I, \angle H, \angle G}$

13. $\boxed{9\text{ cm}}$

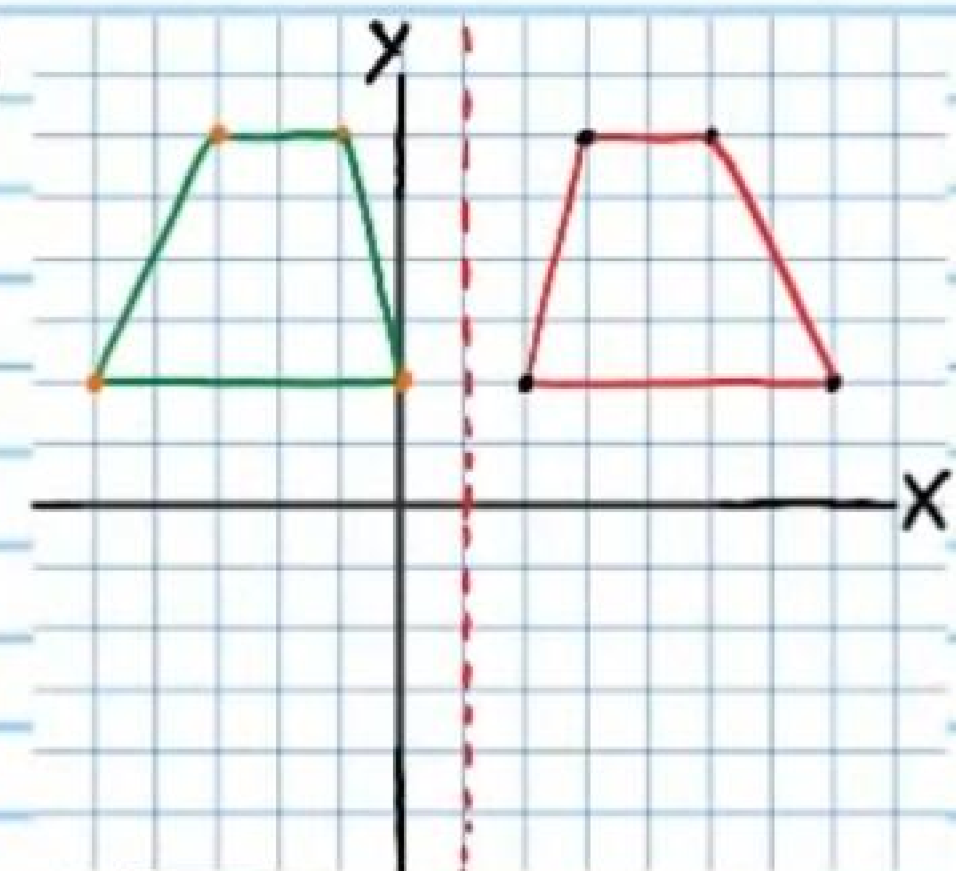
14. $\boxed{\text{No}}$ $\boxed{\frac{3}{4} \neq \frac{5}{7}}$

15. $\boxed{X = 22.5}$

16. $\boxed{1 < X < 17}$

17. An animal is a dog if and only if it has four legs.

18.



19. i) $\boxed{10\text{ ft}}$ ii) $\boxed{16\text{ ft}}$ iii) $\boxed{11\text{ ft}}$ iV) $\boxed{11\text{ ft}}$

20. i) $\boxed{\triangle ABD \cong \triangle ACD \quad \text{SSS}}$ ii) $\boxed{\triangle FEG \cong \triangle IHG \quad \text{ASA or AAS}}$

21. $M_{AC}\left(\frac{1+(-3)}{2}, \frac{3+(-5)}{2}\right) = M_{AC}(-1,-1)$

$M_{DB}\left(\frac{-5+3}{2}, \frac{-2+0}{2}\right) = M_{DB}(-1,-1)$

Therefore, $\overline{AC}$ and $\overline{DB}$ bisect each other.

22

Statement	Reason
I) $m\angle A = 32°$, $m\angle C = 89°$, $m\angle D = 32°$, and $m\angle E = 59°$.	Given.
II) $m\angle A + m\angle B + m\angle C = 180°$	Δ Sum Thm.
III) $32 + m\angle B + 89 = 180$	Subs. Prop of =.
IV) $m\angle B + 121 = 180$	Simplification.
V) $m\angle B = 59°$	Subt. Prop of =.
VI) $m\angle B = m\angle E$ & $m\angle A = m\angle D$	Trans. Prop of =.
VII) $\Delta ABC \sim \Delta DEF$	AA

23

Statement	Reason
I) $\overline{EF} \perp \overline{FG}$ and $\overline{FG} \perp \overline{HG}$	Given.
II) $\angle EFG$ and $\angle HGF$ are rt $\angle$'s.	Def of $\perp$
III) $\overline{FG} \cong \overline{FG}$	Ref. Prop of $\cong$.
IV) $\overline{EG} \cong \overline{FH}$	Given.
V) $\Delta EFG \cong \Delta HGF$	HL
VI) $\angle E \cong \angle H$	CPCTC

24

Statement	Reason
I) $X = 45$	Given.
II) $m\angle DBC = 2X + 4$.	Given (diagram).
III) $m\angle DBC = 2(45) + 4$	Subs. Prop of =.
IV) $m\angle DBC = 94°$	Simplification.
V) $m\angle EDB = 86°$	Given (diagram).
VI) $m\angle DBC + m\angle EDB = 94 + 86$.	Subs. Prop of =.
VII) $m\angle DBC + m\angle EDB = 180°$	Simplification.
VIII) $\angle DBC$ and $\angle EDB$ are supplementary.	Def. of Supp. $\angle$'s.
IX) $a \parallel b$	CIACT

25 $\boxed{\left(-\frac{1}{11}, \frac{25}{11}\right) \text{ or } \left(-\frac{1}{11}, 2\frac{3}{11}\right)}$